

Lect 13B, slide No. 3

$$x(t) = \bar{L}^{-1} \overset{(1)}{(sI-A)} x(0) + \bar{L}^{-1} \overset{(2)}{(sI-A)^{-1}} B U(s) \quad (1)$$

If there is no input  $u(t)$ , the solution is  
No second term in eqn. ~~in (1)~~  $x(t) = \bar{L}^{-1} (sI-A)^{-1} x(0)$

$$\bar{L}^{-1} (sI-A)^{-1} = \phi(t) \text{ is called STM}$$

$$x(t) = \phi(t) x(0) \quad (2)$$

If you know  $\phi(t)$  and  $x(0)$ , you can find state at any time  $t$ .

$$\text{Eqn. (2) is the solution of } \dot{X} = \underline{AX} \quad (3)$$

where  $A$  is a matrix.. ~~if~~

$$\dot{x} = ax \text{ if } a \text{ is scalar}$$

$$\dot{x} - ax = 0 \quad \text{The solution is } x(t) = e^{at} x(0)$$

$$\text{When } A \text{ is matrix } x(t) = e^{At} x(0)$$

$$\phi(t) = e^{At} \quad \text{The properties in slide 4 can be derived from this}$$

In the last tutorial sheet, there are problems to find the nature of eqbm points. There are nonlinear Eqns.

See lecture 13B, slide 10  $\ddot{\theta} + \sin \theta = 0 \quad \dot{x}_1 = x_2 = f_1$   
Nonlinear. Jacobian  $J$  linearizes the system  $\dot{x}_2 = -\sin x_1 = f_2$   
about the eqbm points.

eqbm points  $(0,0)$  and  $(\pi,0)$

Find  $J$  at  $(0,0)$  = eigenvalues

$J$  at  $(\pi,0) \rightarrow$  unstable

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

